Binfei Ji

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Education

Georgia Institute of Technology — Atlanta, GA
Master of Science in Computer Science (Machine Learning Track) — *Aug 2025 – Dec 2026 (Expected)*

Peking University, Yuanpei College — Beijing, China
Bachelor of Science in Physics — *Sep 2017 – Jul 2021*

Professional Experience

Adobe — San Jose, CA
Research Intern — *May 2026 – Present*

- **Speaker-Aware Audio-Visual Captioning:** Fine-tuned Qwen3-Omni to predict **who spoke what and when**, along with emotion and delivery, improving **speaker-attribution by 12 points** over the base model.
- **Data & Benchmark Pipeline:** Built an internal benchmark and curated a **200K-clip** training corpus using diarization, active-speaker, and A/V-sync models for quality filtering, followed by stratified resampling across language, duration, speaker count, and shot count; used Gemini-assisted annotation for transcripts and speaker descriptions.
- **Cross-Modal Binding Diagnosis:** Identified **audio-visual speaker grounding** as the primary bottleneck: conditioning on reference transcripts and timestamps substantially outperformed end-to-end generation, motivating a decomposed training strategy that separates audio perception from cross-modal binding.

Eigen AI — Remote
Research Intern — *Aug 2025 – Mar 2026*

- **Structured Visual Grounding:** Reframed end-to-end accident video understanding with explicit vehicle tracks and depth-based collision candidates, providing Qwen3-VL-235B with grounded object identities and spatial cues; improved accident Acc from **∼20% to 82%** and involved-vehicle F1 to **74%**.
- **Verifiable RL:** Observed involved-vehicle **pass@8 reaching ∼90%**, revealing substantial latent capability beyond single-sample performance; built a **5K-video** RL dataset with directly verifiable rewards and trained Qwen3-VL-4B with DAPO, improving accident F1 from **40% to 88%** and involved-vehicle F1 from **15% to 78%**, **surpassing the zero-shot 235B baseline**.

XPENG Motors — Beijing, China
Senior Machine Learning Engineer, Foundation Models and AI Infrastructure Team — *Sep 2023 – Jun 2025*

- **VLM-Based End-to-End Driving:** Contributed to the multimodal training framework for XPENG's end-to-end autonomous-driving program, enabling VLM-based trajectory planning from visual and driving-context inputs.
- **Geo-Visual Fusion:** Designed a visual-prompting strategy that rendered BEV representations into the VLM input, injecting explicit road geometry into trajectory reasoning and reducing lane-boundary violations.
- **Sparse BEV Perception (PETR):** Implemented sparse query-based detectors (PETR series), reducing inference latency by **20%** through object-centric sparse computation.

XPENG Motors — Beijing, China
Machine Learning Engineer, Perception Team — *Aug 2021 – Sep 2023*

- **Profile-Guided Network Optimization:** Reduced BEV model latency by **50%** on NVIDIA Orin through profile-guided architectural optimization, including LayerNorm-to-BatchNorm substitution, decoder simplification, and relocating ConvLSTM to lower-resolution feature maps.

Publications

AgentSuite: Toward More Reliable Agent Evaluation with a Component-Based Benchmark Auditing Pipeline
Hyewon Suh*, **Binfei Ji***, et al. (* Equal Contribution) — *ICML 2026*

BlockBatch: Multi-Scale Consensus Decoding for Efficient Diffusion Language Model Inference
Xiaoyou Wu, Cheng-Jhih Shih, **Binfei Ji**, et al. — *Findings of EMNLP 2026*

Technical Skills

Languages & Frameworks: Python, C/C++, PyTorch, TensorRT, vLLM, SGLang, verl, MMDetection, Bash